

































Cy-Ramick plate — hand-wired, no PCB.

Power: 4xAA alkaline (6.0–6.4 V fresh) -> holder switch -> D1 -> ESP32 VIN (~5.3 V)
VIN -> D2 -> DFPlayer VCC (~4.5 V; DFPlayer is rated 3.2–5.0 V).
D1 also blocks USB 5 V from back-feeding the cells while programming.
The ESP32's onboard AMS1117 makes 3V3 for the GY-521 and all five TCRT5000 modules
(ESP32 GPIOs are not 5 V tolerant).
DFPlayer RX needs no series resistor: the ESP32 is 3.3 V logic.
C1 is optional (DNP): fit it across DFPlayer VCC/GND if the player resets or pops at high volume.
Buttons: any spare even GPIO (0, 12, 14, 18) acts as button 1 and any spare odd GPIO
(5, 13, 15, 19, 23) as button 2. SW2/SW3 show the main wiring on GPIO18/GPIO19.
Avoid the strapping pins 0, 2, 5, 12, 15 for anything that pulls them at power-up.

Hand-wired point to point; generated by make_schematic.py

KnightHack 2026

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File: plate.kicad_sch

Title: Cy-Ramick: The Sentient Plate

Size: A3	Date:	**Rev: 2**
KiCad E.D.A. 10.0.7		Id: 1/1